House Price Prediction Using Machine Learning

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Insights and Summary

- **Key Price Drivers:** The correlation analysis revealed that *area* ($r = 0.54$) and *bathrooms* ($r = 0.52$) were the strongest predictors of house prices. Other important factors included air conditioning ($r = 0.45$), number of stories ($r = 0.42$), and parking availability ($r = 0.38$).
- **Model Performance:** The Linear Regression model achieved an R^2 score of approximately 0.65, indicating that it was able to explain about 65% of the variation in house prices. The Actual vs. Predicted plot shows a reasonably strong linear relationship, although prediction errors increase for higher-priced properties.
- **Price Distribution:** The histogram indicates that most houses are priced between 3 million and 6 million. The distribution is positively skewed, with a smaller number of premium properties priced above 10 million.
- **Interesting Observation:** Features such as guest rooms and basements exhibited relatively weak correlations with house prices compared to core structural features like area and bathroom count. This suggests that buyers place greater value on usable space and essential amenities.
- **Business Recommendation:** Real estate developers and property investors should focus on increasing living area, improving bathroom facilities, and providing modern amenities such as air conditioning and parking. These features have the greatest impact on property valuation and can significantly improve market value.

Project Structure

```
HousePricePrediction_ChiragSharma
|-- analysis.ipynb
|-- Housing.csv
|-- summary.pdf
‘-- charts
    |-- chart1_price_distribution.png
    |-- chart2_correlation_heatmap.png
    ‘-- chart3_actual_vs_predicted.png
```